REPORT — Bridge C order4b (weight-0 law for A2)

PREREG sha256 5346382c…dbef2 (unchanged). Script hashes in scripts/SHA256SUMS. Category B: numerical ranks. Mod-p certification is not feasible because the entries are transcendental.

Anchor

A012 at x=1 reproduces the order3c A2 value exactly (difference 0.0). **Pass.**

Sign control (gating)

- A0 at w=1: rank 36, closed.
- A0 at w=0: joint rank 72, not closed.
- **Passed.** The test separates weights.

Tests (105 companions × 216 points)

test	rank	joint	closed
T0: A2, w=0	73	146	**no**
A2, w=½ (descriptive)	73	146	no
A2, w=1 (descriptive)	73	146	no
T1: affine span [A2, |x|^{½}A1, |x|A0]	214	216	no

Verdicts

- **T0:** the weight-0 law is falsified (Category B). The S-image of A2 is linearly independent of A2 (joint rank = 2 × 73).
- **T1:** not closed. The span rank is 214 out of 216 rows, so the test is close to saturation and its discriminating power is low. It is reported as stated, not as strong evidence.

order4b notes (descriptive)

- **Rank of A2.** A2 has rank 73, between the rank of A0 (36) and that of A1 (105). This is consistent with the B1·B1 and B2 finite sums adding new directions on top of the area and drift terms.
- **Pure law.** The doubling of the joint rank at every weight rules out a pure vector law at order t^0 on this grid.
- **Affine law.** The affine test is near saturation. A sharper version needs more rows: more points XS (denominators beyond 15), or the test split by characteristic block (24 + 12 for A0).
- **Process.** Eight point-chunks were run in parallel processes, each single-threaded under nice, each under 2.5 GB.